

SHAPEIC: DESIGN-SPACE EXPLORATION

*Hierarchical LUT-based design-space exploration
for analog IC sizing*

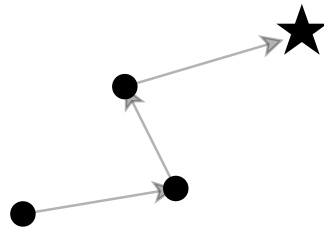
Daniel Arevalos

Munich University of Applied Sciences HM

Automatic Analog Sizing

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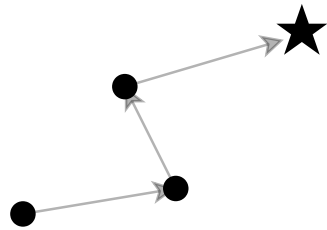
► **Optimization-driven**



- ✓ Strategic evaluation of candidates
- ✓ Many algorithms available

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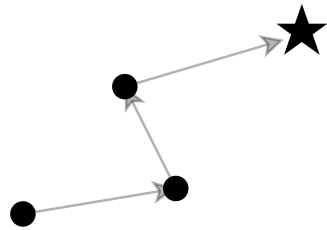
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- ✓ Strategic evaluation of candidates
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- ✗ Partial view of the design space
- ✗ May be sensitive to local optima

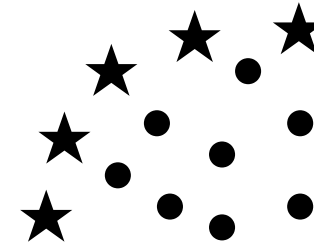
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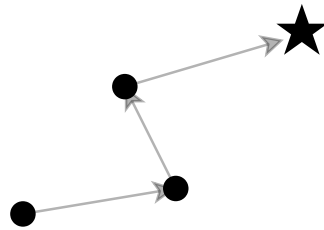
► Design-space exploration



- ✓ Design insight
- ✓ Alternative solutions

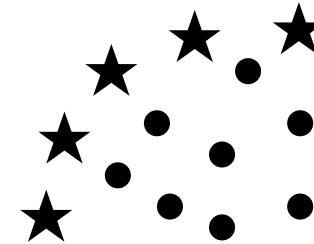
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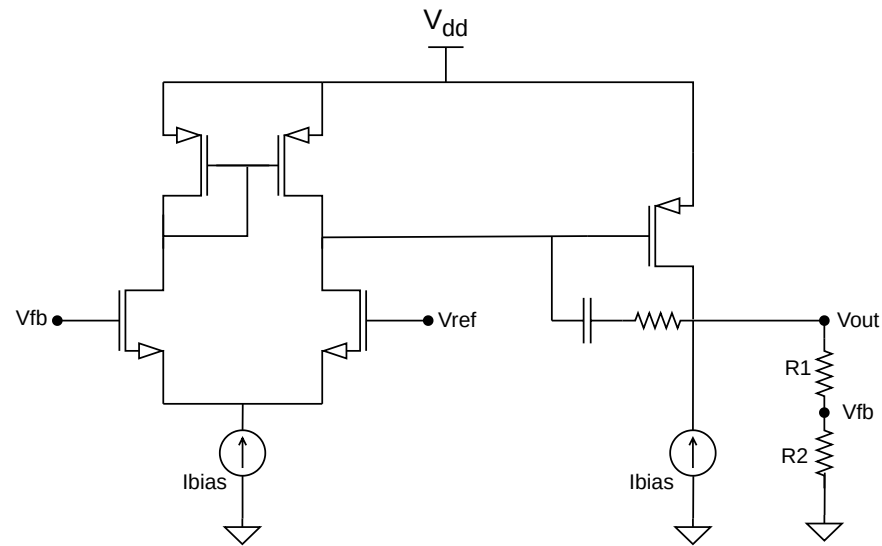
► Design-space exploration



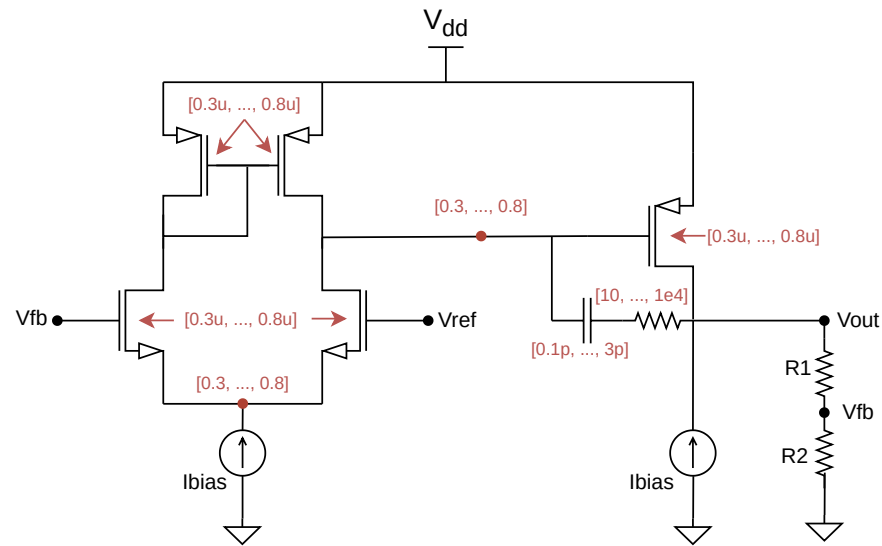
- ✓ Design insight
- ✓ Alternative solutions
- ✗ Expensive to build
- ✗ Scales poorly without abstraction

Let's explore a 2-Stage Amplifier

- ▶ Start with a simple 5-transistor topology

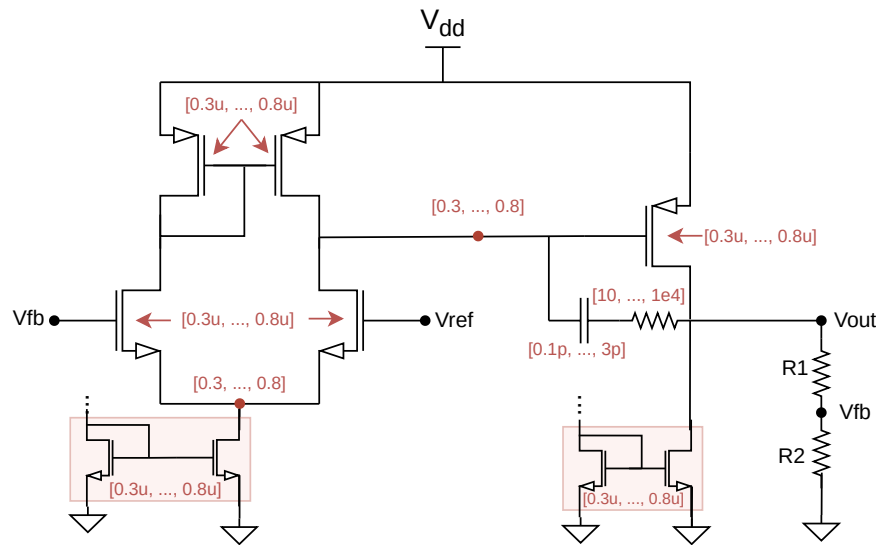


Let's explore a 2-Stage Amplifier



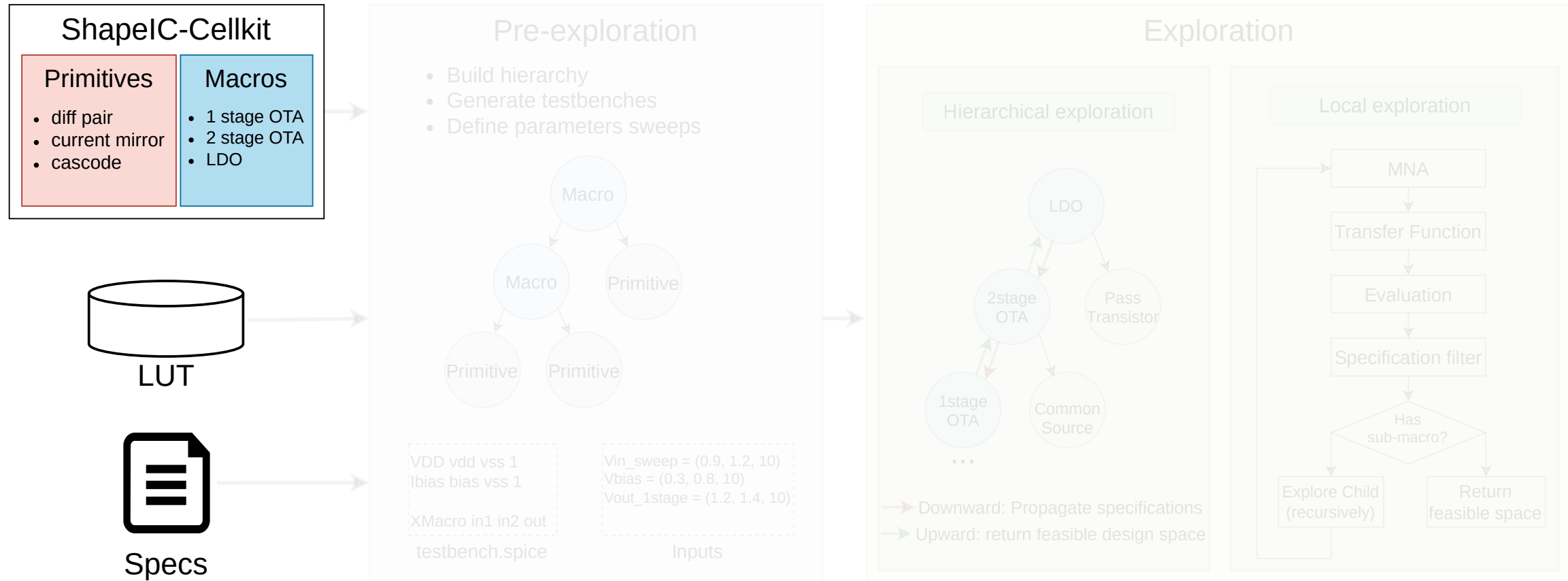
- ▶ Start with a simple 5-transistor topology
- ▶ 1250000 design candidates
 - ▶ Simulation: ≈ 2 h
 - ▶ Equation evaluations: ≈ 2 s (excluding manual equation derivation)

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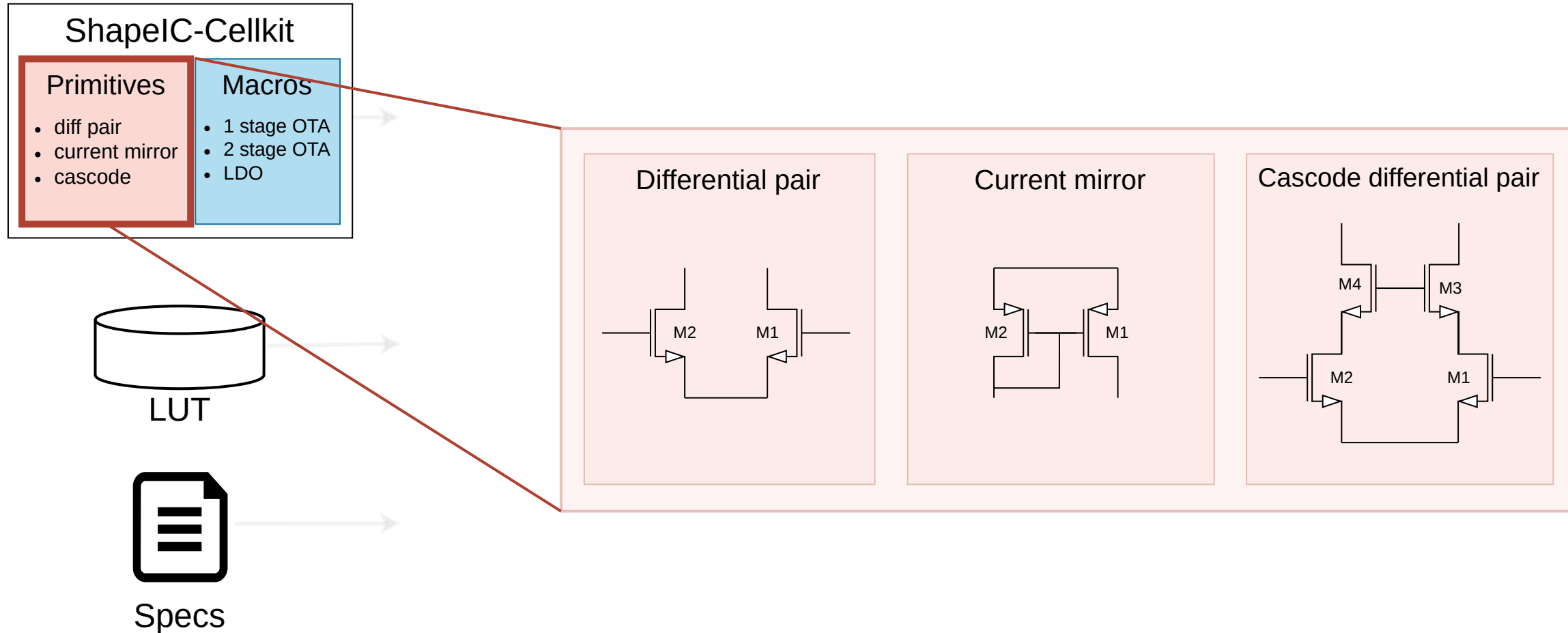


- ▶ Start with a simple 5-transistor topology
- ▶ 1250000 design candidates
 - ▶ Simulation: ≈ 2 h
 - ▶ Equation evaluations: ≈ 2 s (excluding manual equation derivation)
- ▶ 31250000 design candidates
 - ▶ Simulation: ≈ 3 days
 - ▶ Equation evaluations: ≈ 1 min (new model/equations required)

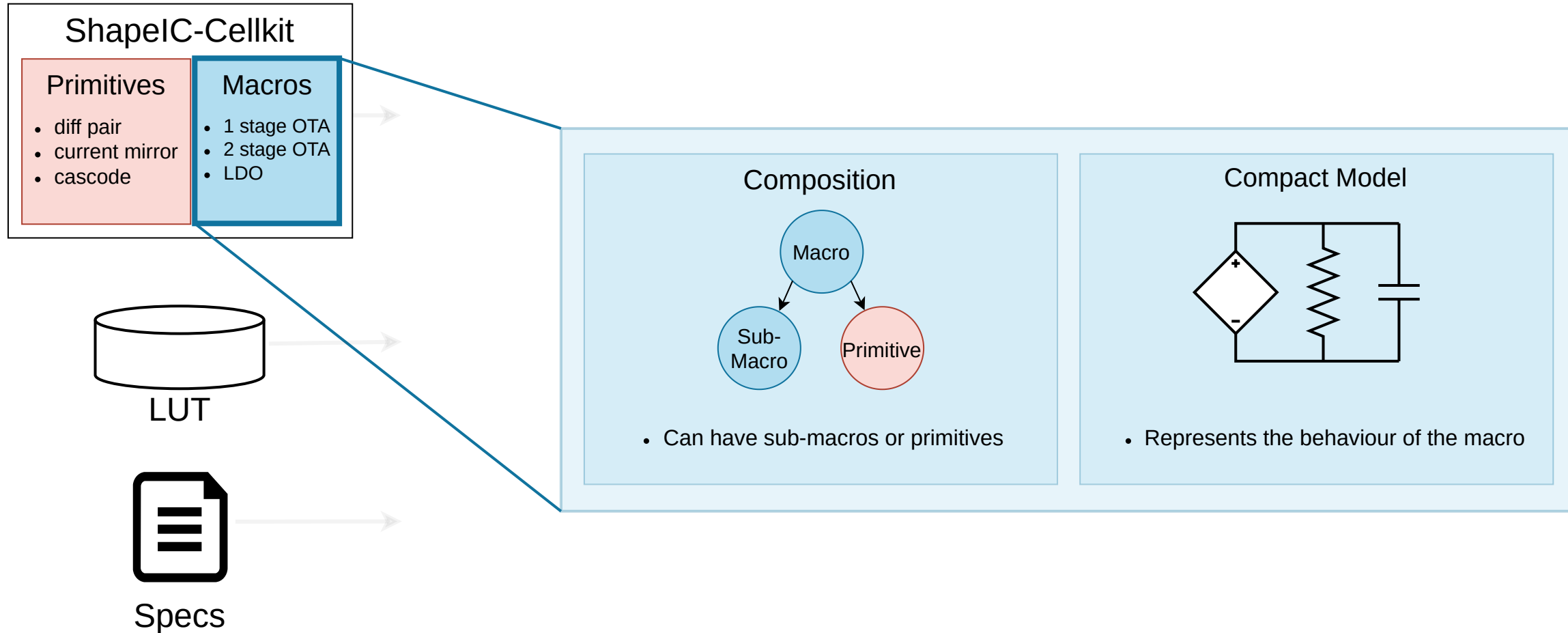
ShapelC: Building the Design Space



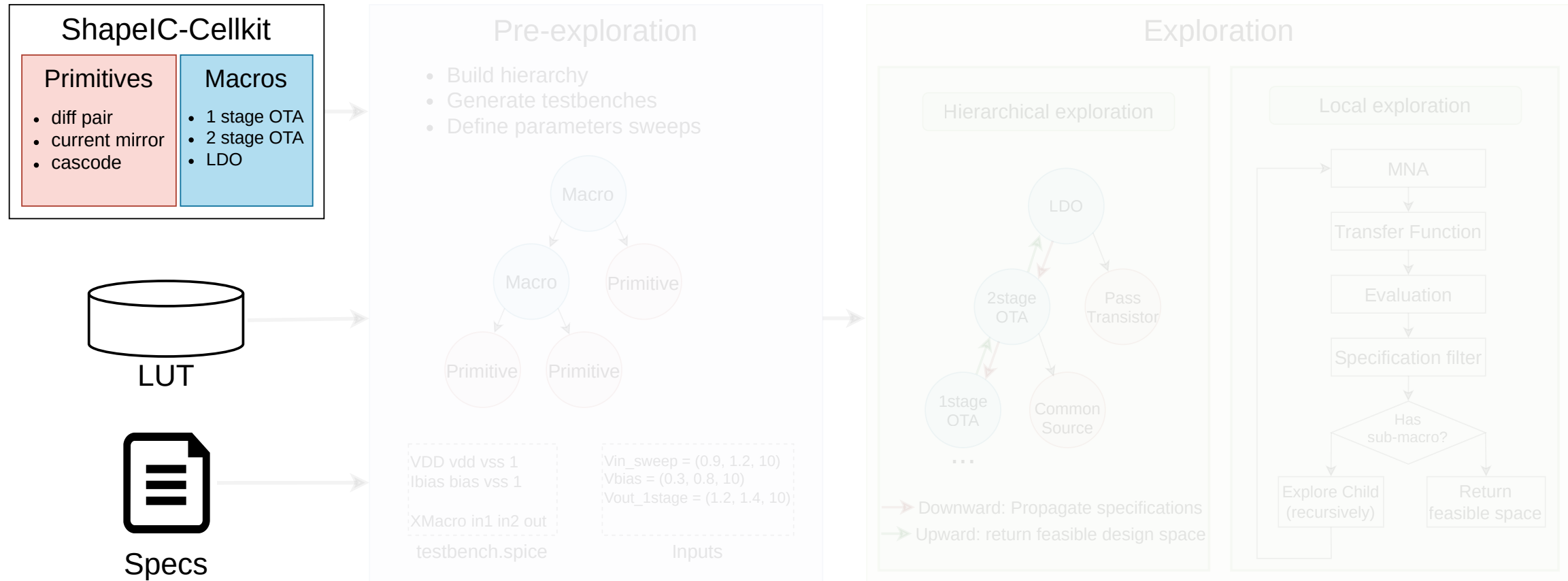
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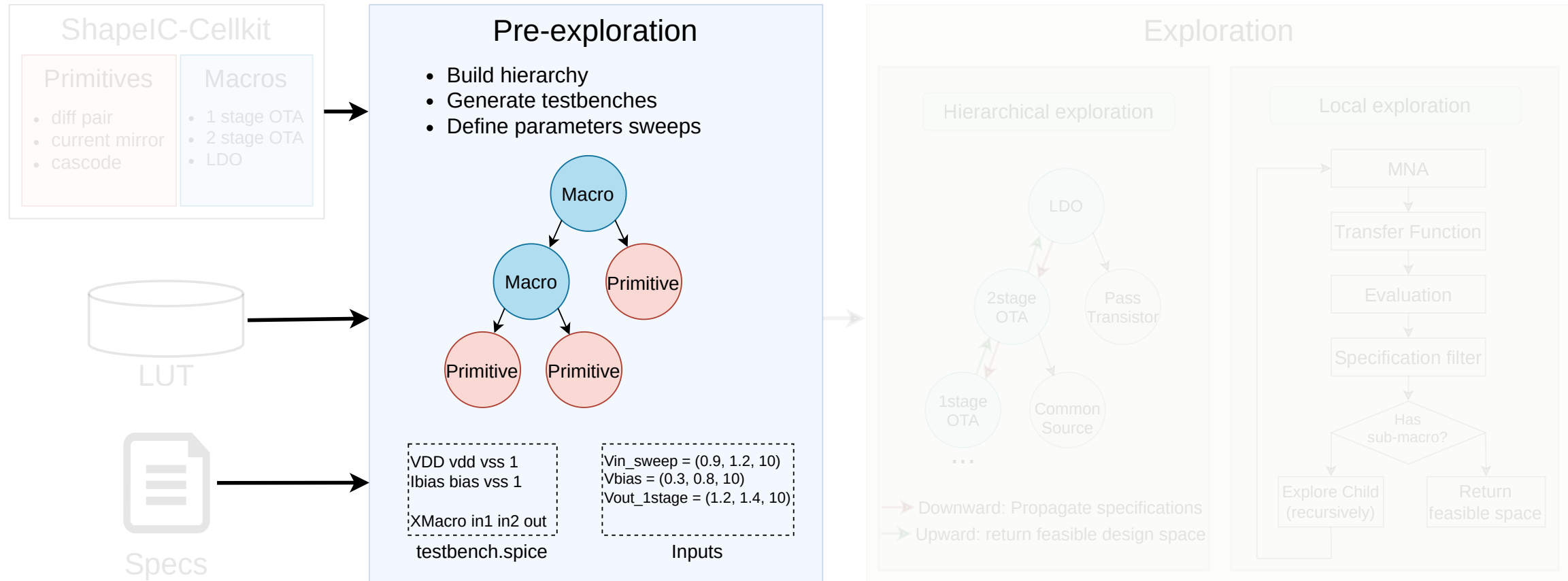
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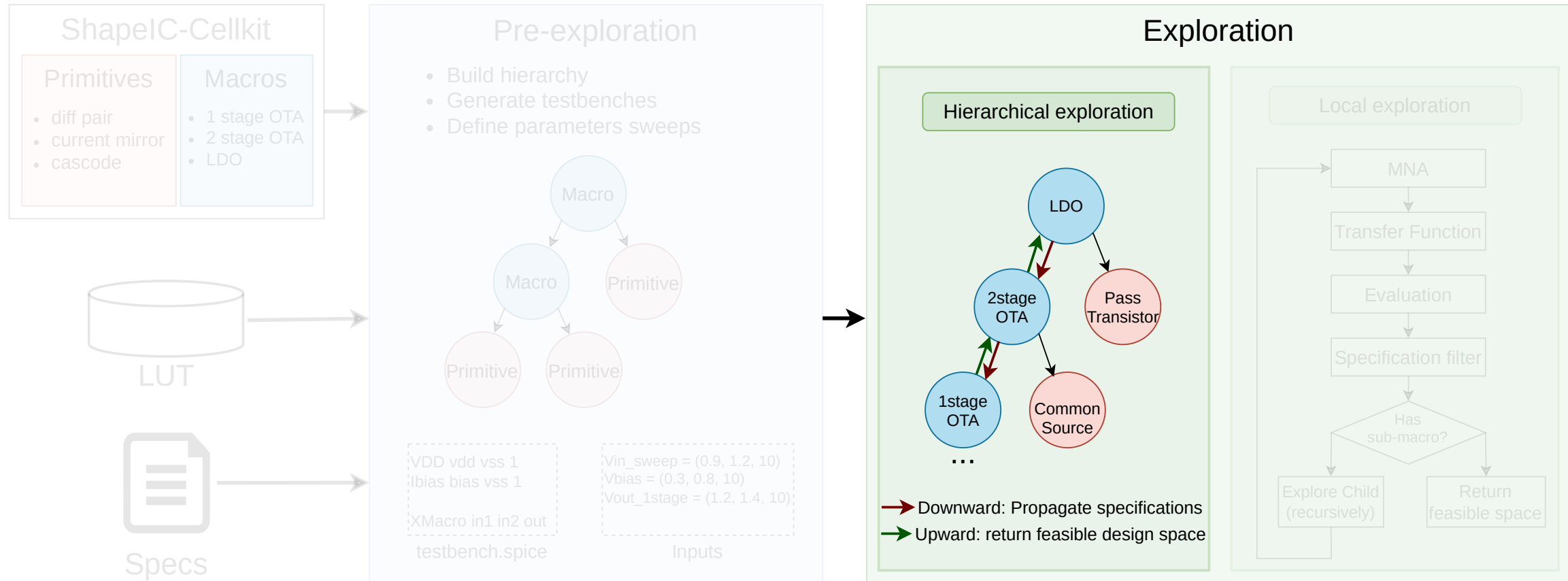
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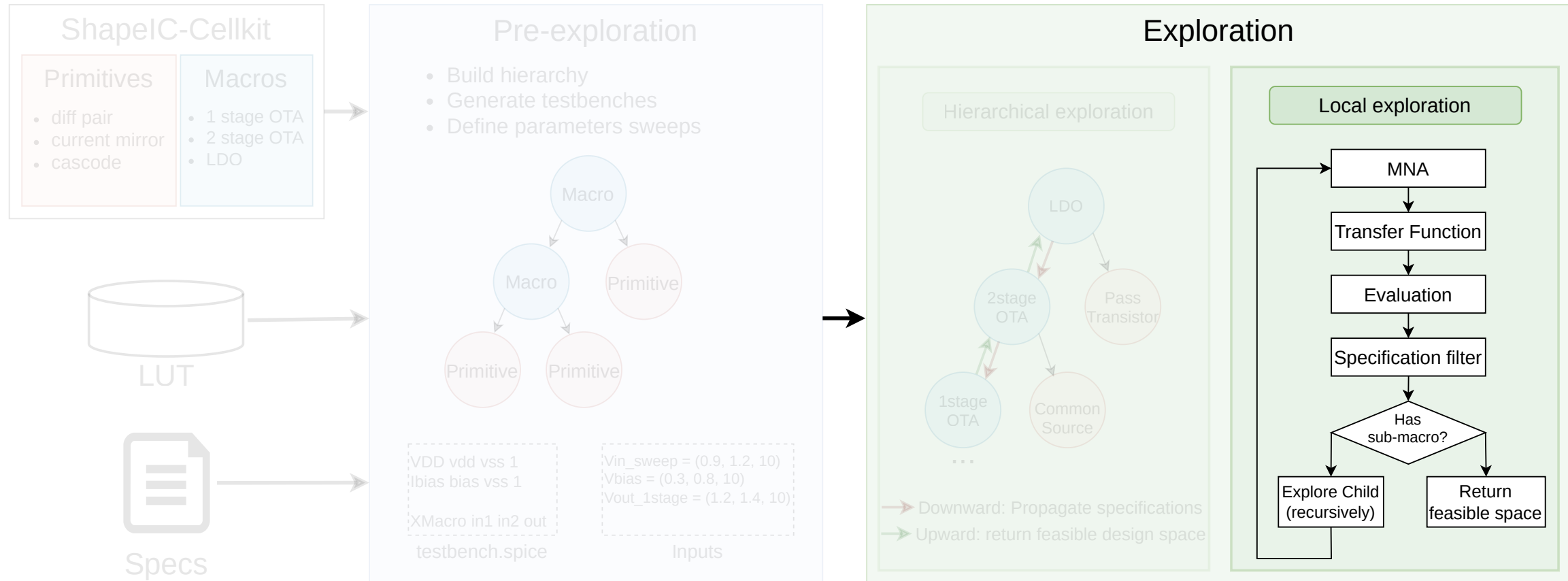
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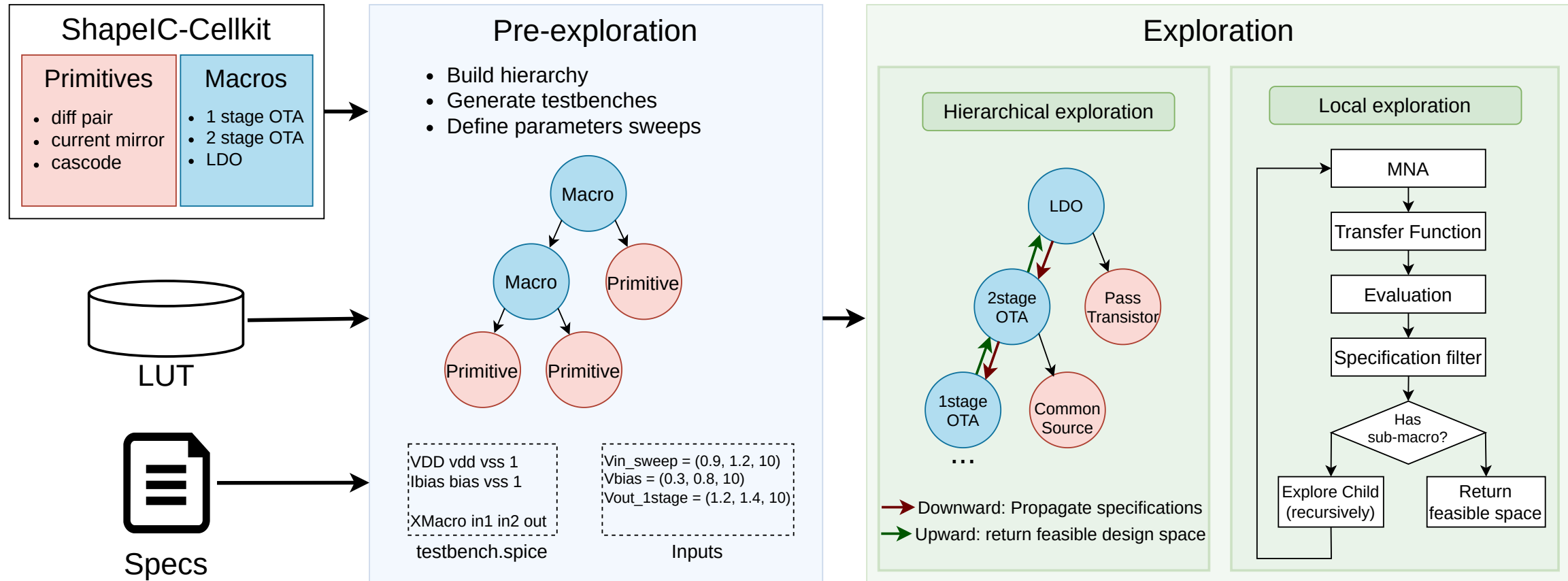
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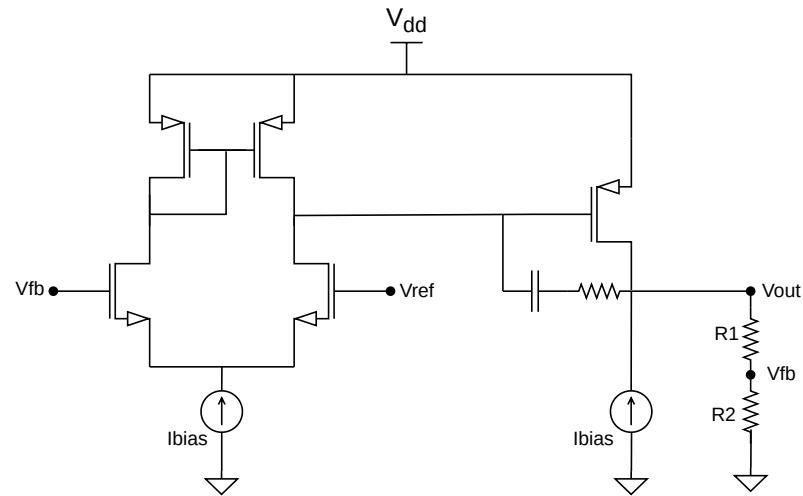
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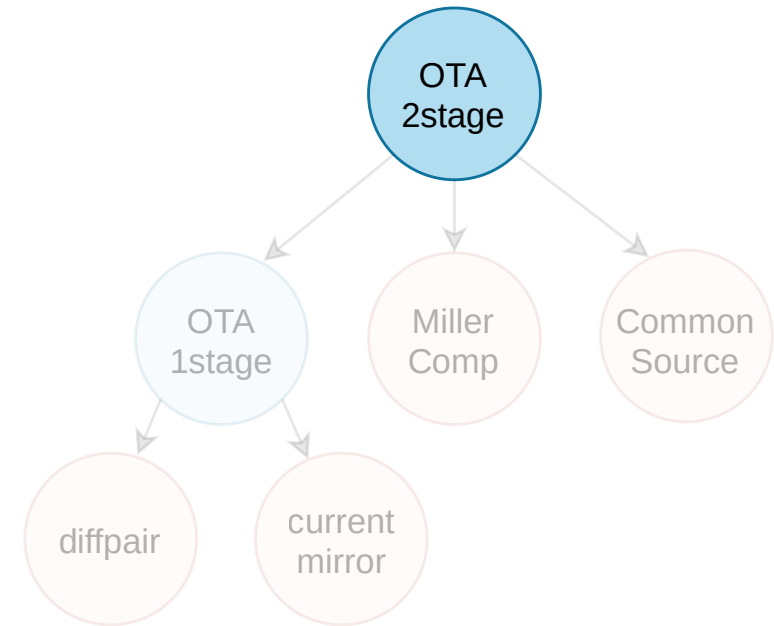
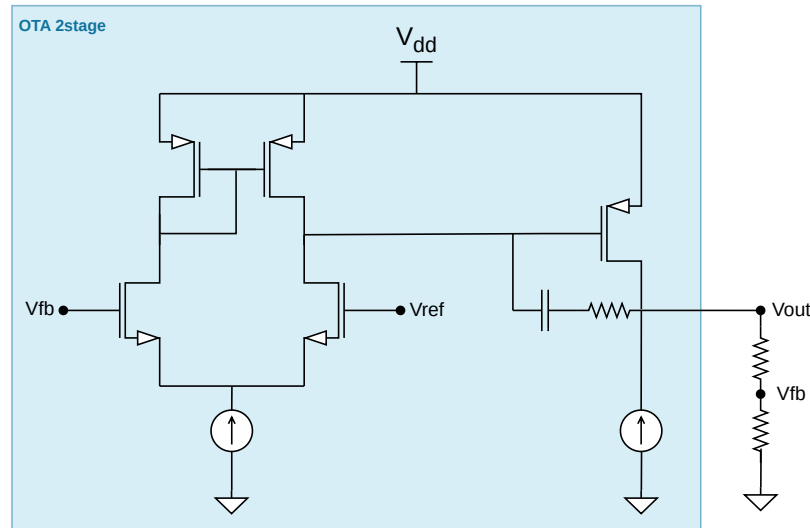
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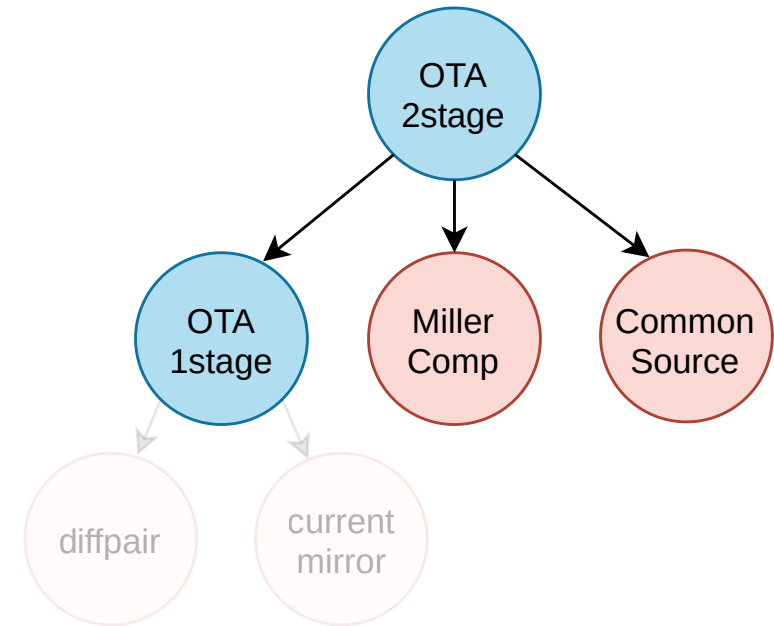
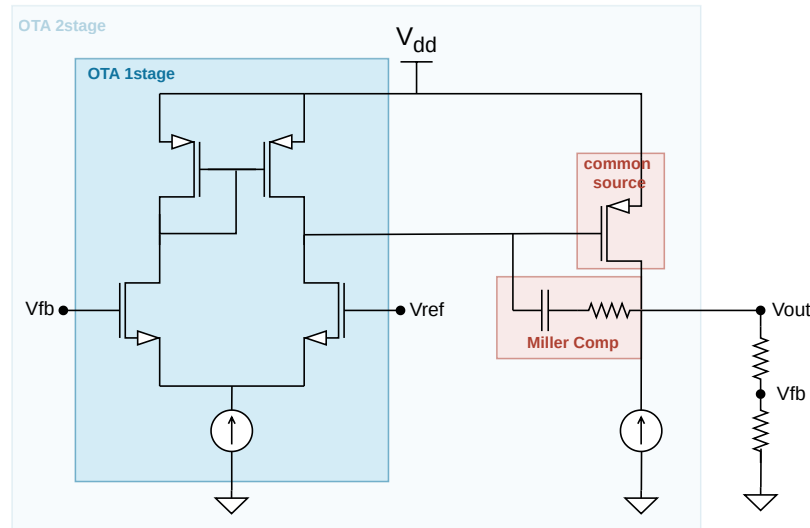
ShapelC in Action: 2 Stage Amplifier (OTA)



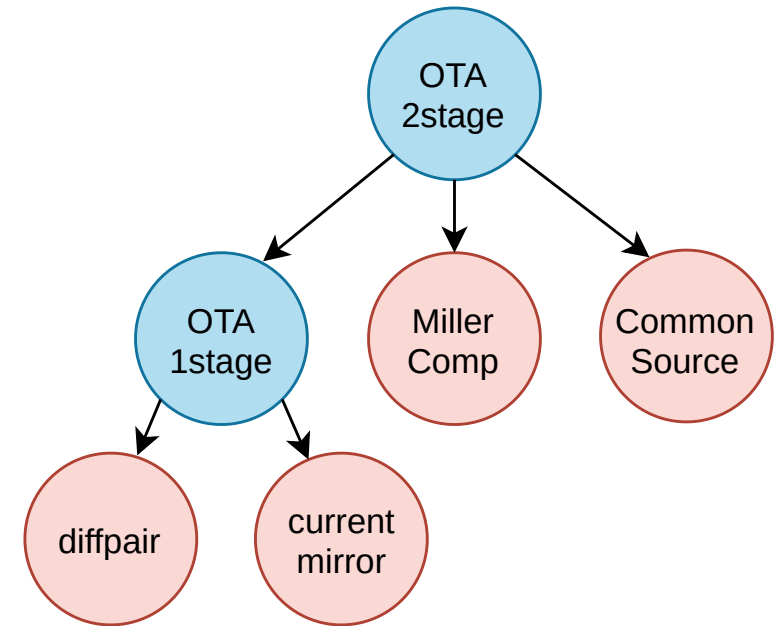
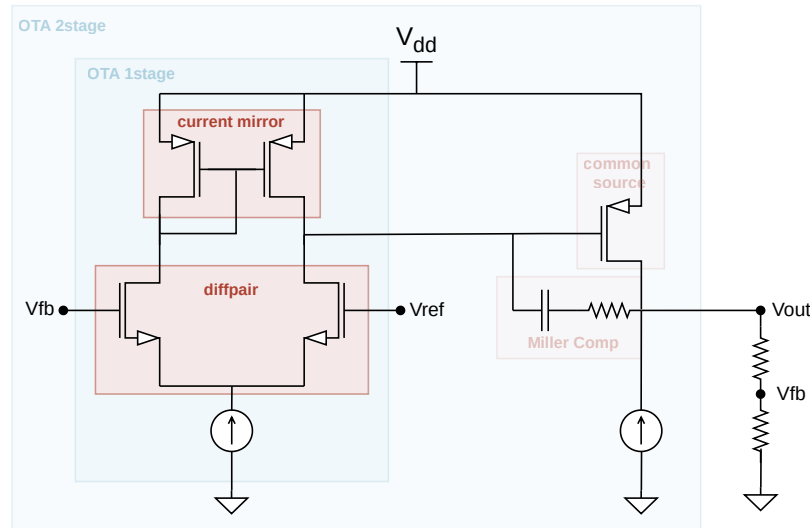
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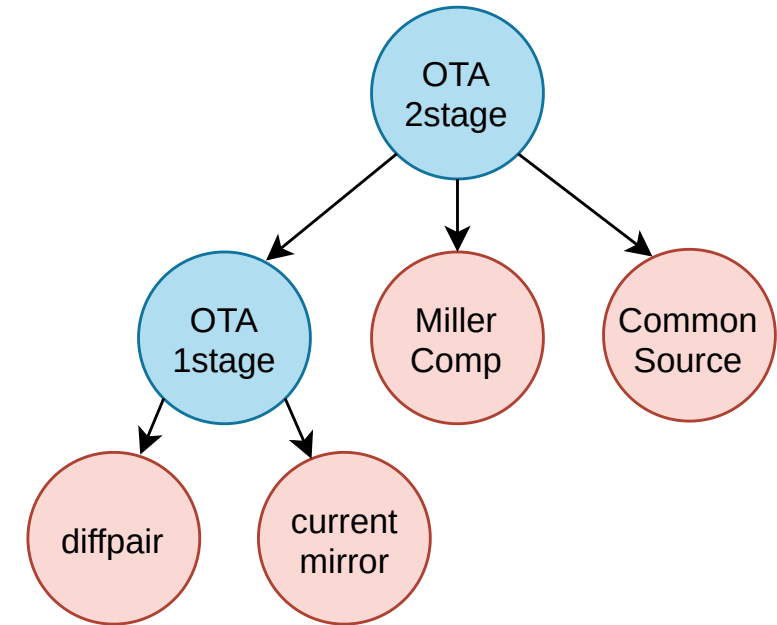
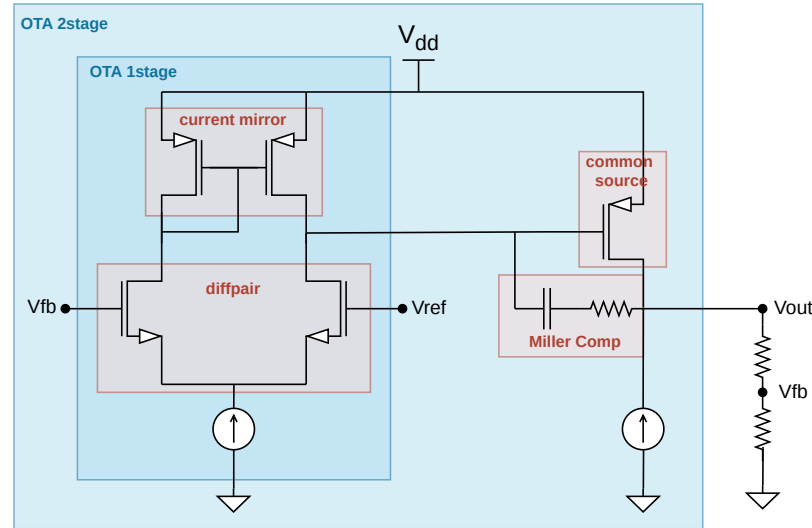
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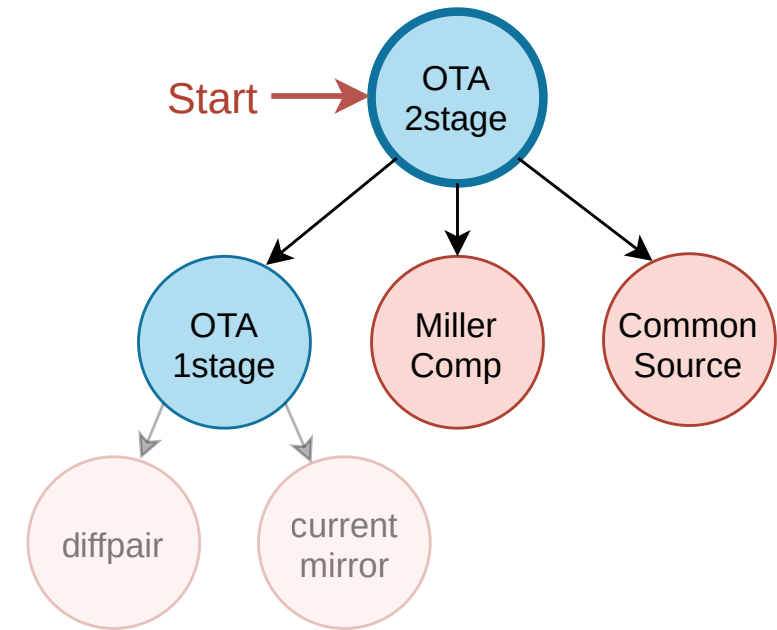


- ▶ Testbench: AC analysis
- ▶ Specs: $Gain \geq 70dB$, $BW \geq 10kHz$, $PM \geq 60^\circ$

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1. Hierarchical exploration

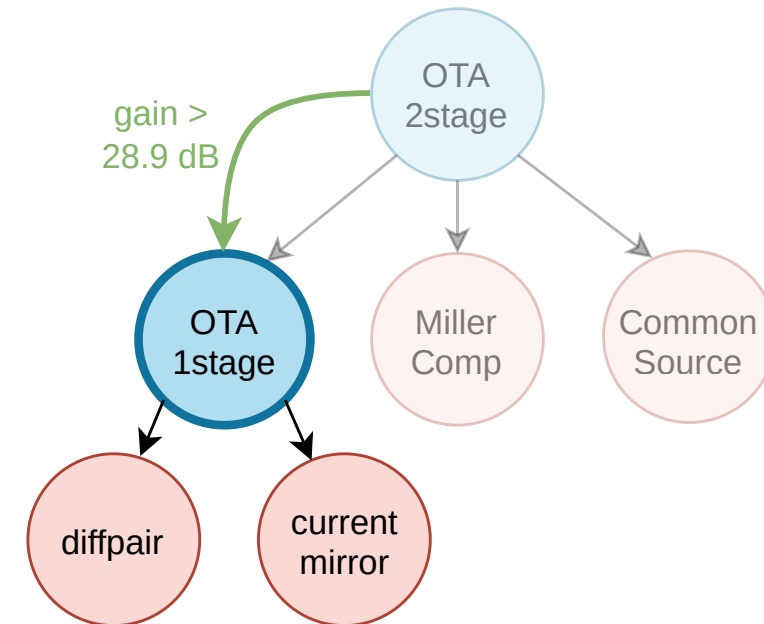
- ▶ TOP-level preview: 500k → 2021
- ▶ Derived: 1stage *Gain* ≥ 28.89 [dB]



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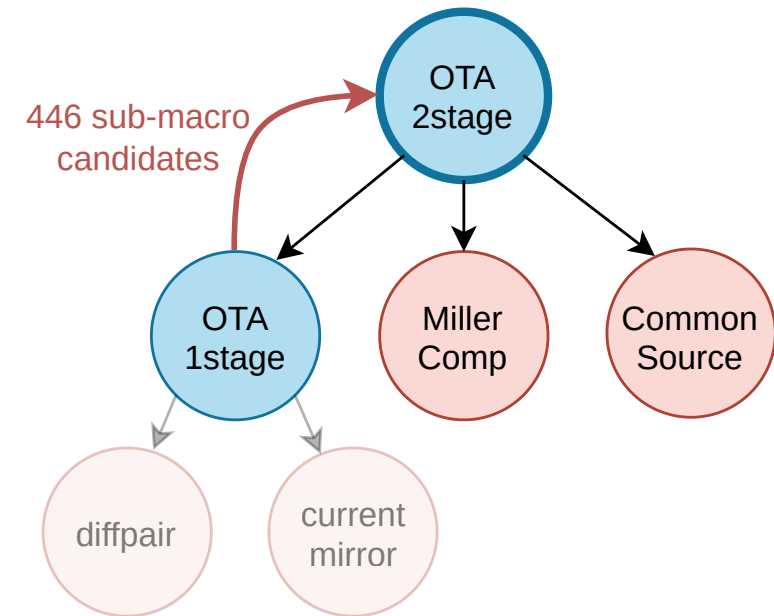
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- ▶ 1stage OTA exploration: 1050 → 446
 - ▶ 446 feasible candidates returned



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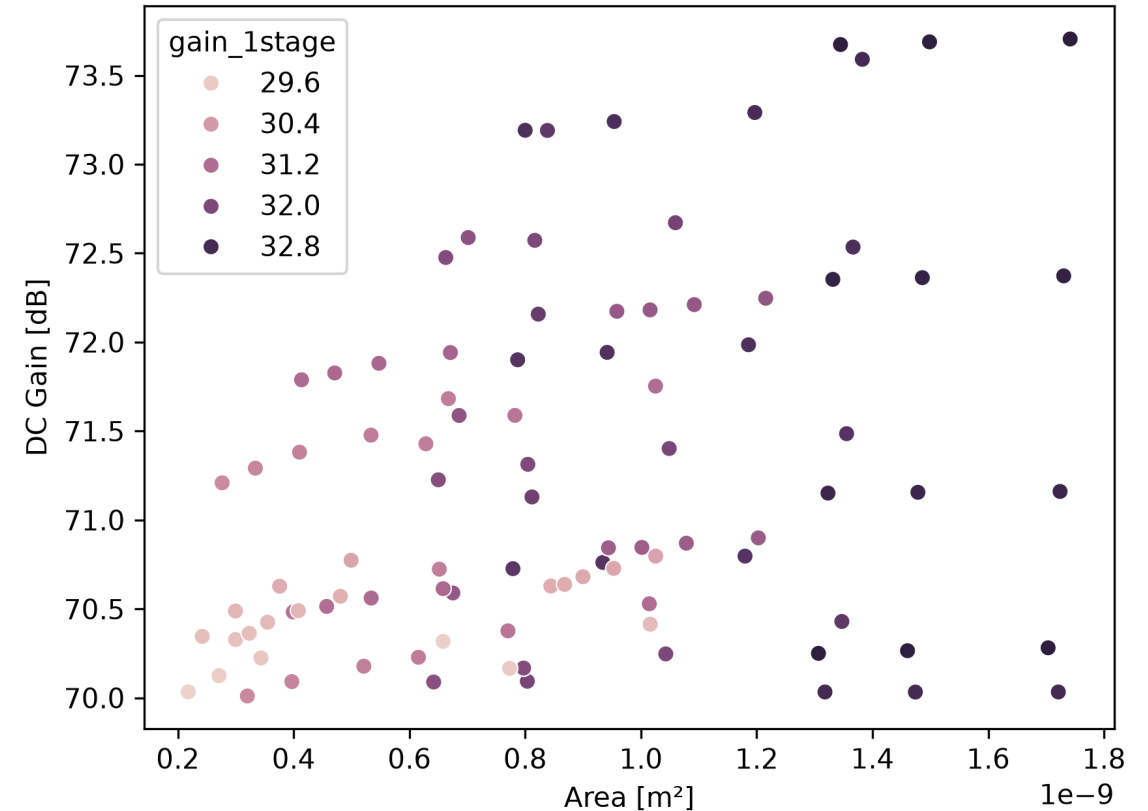
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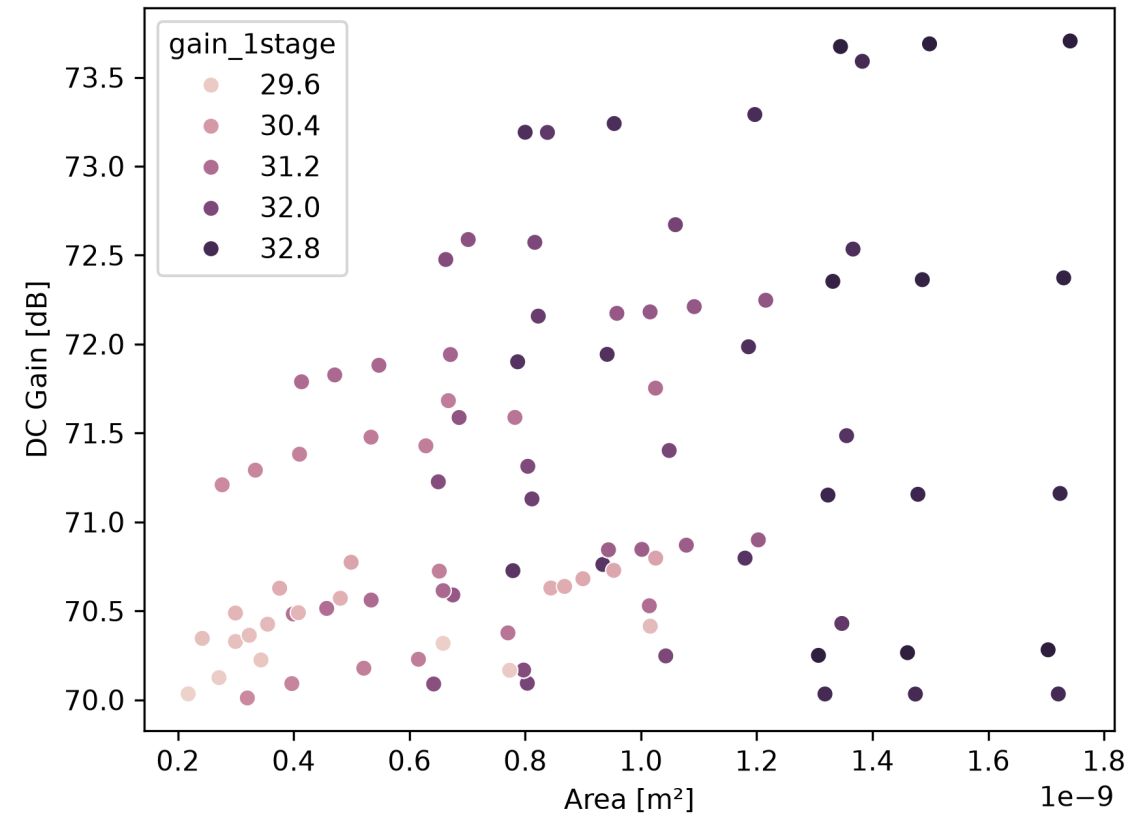
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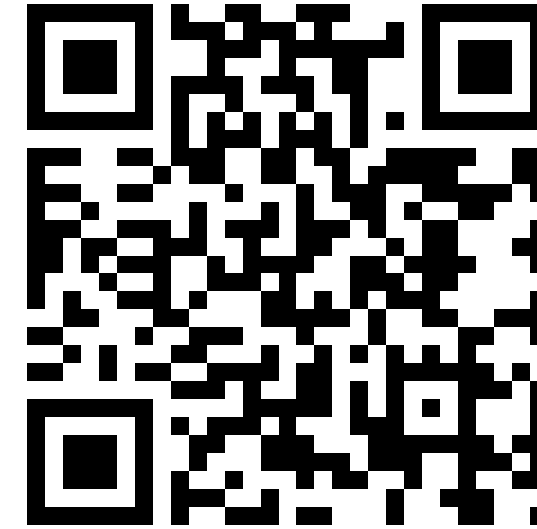
2. Runtime

- ▶ Exploration only: **2.79 s**
- ▶ Full run incl. LUT load: **8.38 s**



What is Next for ShapeIC?

- ▶ Analysis
 - ▶ DC Analysis
 - ▶ Transient analysis
- ▶ Design & Exploration
 - ▶ Interactive GUI
 - ▶ Layout-aware exploration
- ▶ ShapeIC-Cellkit
 - ▶ More primitives
 - ▶ More macros



github.com/ShapeIC/shapeic

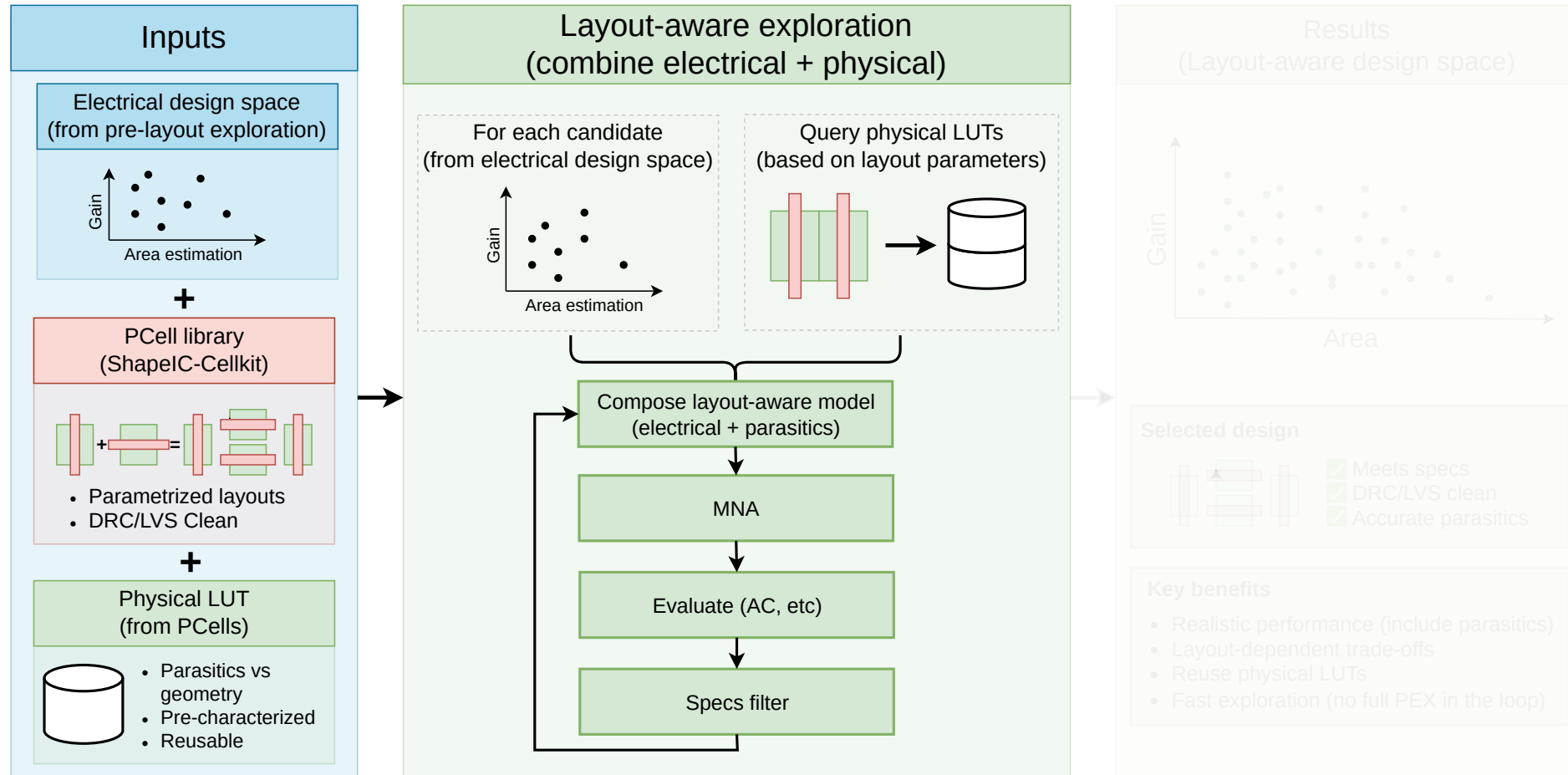


Supported by OCDCPro

Layout-Aware Exploration



Layout-Aware Exploration



Layout-Aware Exploration

